

Conformal Prediction

risk control

testing / multiple testing

outlier detection

online learning

generative models

llm ✓ Agentic

History:

1995 Vladimir Vovk, UCL

Vladimir Vapnik, SVM

Alexander Gammerman

2000 Larry Wasserman CMU

Lei Jing

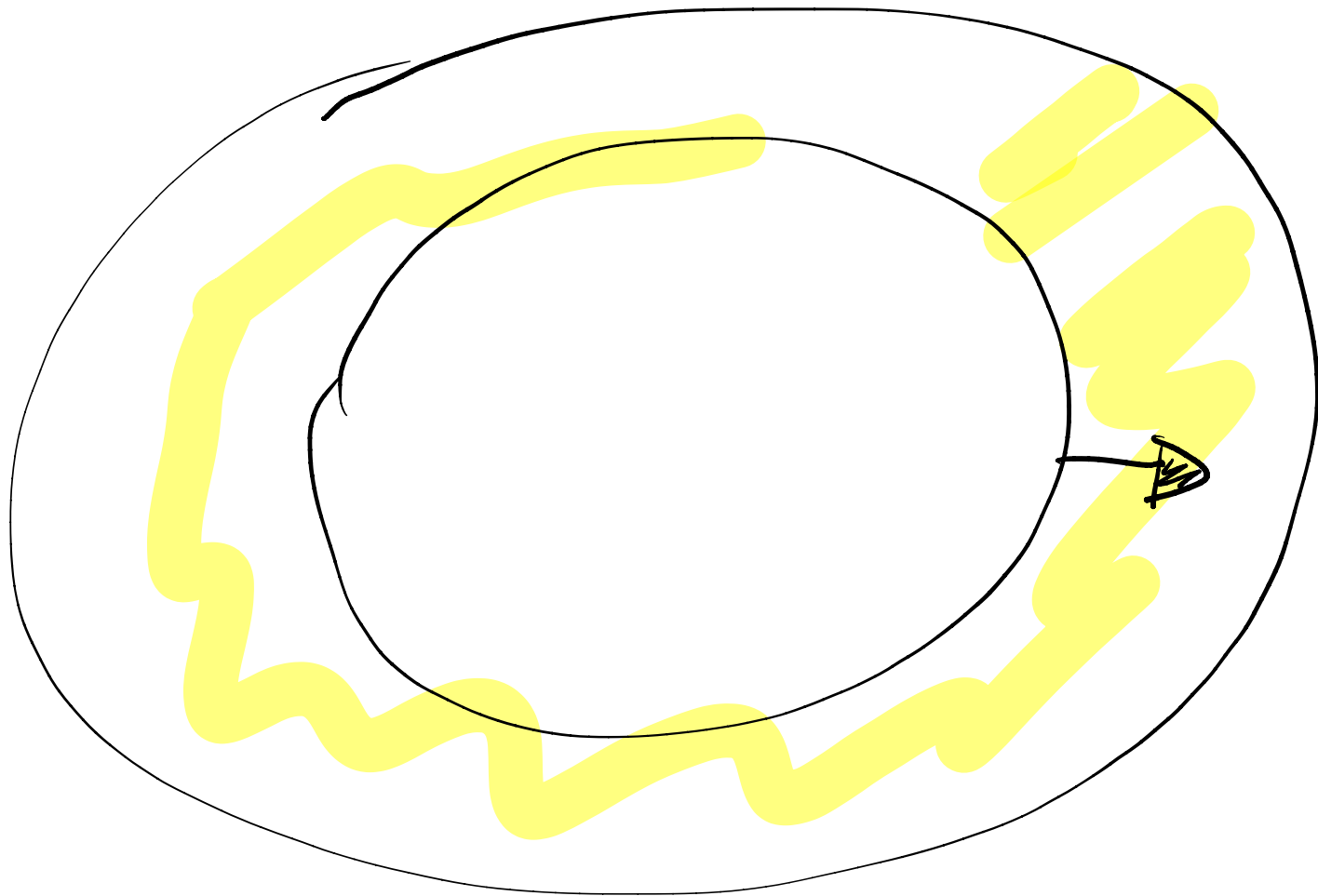
2023 Keynote Emmanuel
Candes

UQ important ?

Lao Zi

知不知，上，不知知，病
↓

AI,



Can eliminate Uncertainty?

Other domain?

Other data?

error ✓

1. Distribution shift.

$D_1 \xrightarrow{\text{Train}} \text{LLM}, \quad \text{LLM} \xrightarrow{\text{Apply}} D_2$

$D_1 \neq D_2$

2. Data, finite

$$n \rightarrow \infty$$

3. $\|\hat{f} - f^*\| \neq 0$

Why UQ is important?

Medical

Legal Judicial

Financial

QQ

①

Bayesian

Computational cost
prior specification

②

Quantile regression

Model
specification

③

Statistical error (closed form)

Strong assumption

④

Modern: Conformal Pred,

PCS (bin xu)

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Bootstrap

computation
cost

CP: finite sample guarantee ✓

Model-free ✓

Data exchangeable (iid) ?

→ Covariate shift ✓

distribution shift ✓

Quick Tutorial:

Objective: (standard)

$$(X_i, Y_i) \sim P, \quad i = 1 \dots n, \text{ iid.}$$

$X \in \mathbb{R}^p$ feature

Y outcome

discrete

continuous

To obtain a prediction set $\hat{C}_n$

such that for a new iid pair

$$(X_{n+1}, Y_{n+1}) \sim P,$$

$$\mathbb{P}(Y_{n+1} \in \hat{C}_n(X_{n+1})) \geq 1 - \alpha$$

α - user specified $\in (0, 1)$

$$\hat{C}_n = (-\infty, \infty)$$

$|\hat{C}_n|$ is small ✓

Discrete: $\mathcal{Y}_{n+1} = \{\text{fox}\}$

$$\hat{C}_n = \{\text{fox, wolf, cat, dog}\}$$

How we do it ?

- Given a prediction model $f(\cdot)$

$$\hat{f}(x) \rightarrow \hat{y} \neq y$$

- D_1 . (proper) training set

D_2 . calibration set

D_{Test} test set

$X_0, \textcircled{Y_0}$

unknown

- $D_1 \cap D_2 = \emptyset$

$$n_1 = |D_1|, \quad n_2 = |D_2|$$

Step 1: $D_1 \xrightarrow{\text{train}} \hat{f}_{n_1}$

Step 2: Compute residuals (absolute)

$$R_i = |y_i - \hat{f}_{n_1}(x_i)|$$

for $i \in D_2$

$$\hat{q}_{n_2} = \lceil (1-\alpha)(n_2+1) \rceil \text{ smallest of}$$

$$R_i, \quad i \in D_2$$

$R_1, R_2, \dots, R_{D_2}$

↓ Rank

$R_{(1)}, R_{(2)}, \dots, R_{(D_2)}$

↑
 $k = \lceil (1-\alpha)(n_2+1) \rceil + h$ of R_i

$1-\alpha = 0.25$, $n_2 = 9$, $k = 3$

Step 3 :

$$\hat{C}(x) = \left[\hat{f}_{n_1}(x) - \hat{g}_{n_2}, \hat{f}_{n_1}(x) + \hat{g}_{n_2} \right]$$

Statistical Guarantee!

$$P(Y_0 \in \hat{C}_n(X)) \geq 1 - \alpha$$

How it works?

Simplified Problem:

$$\underbrace{Y_1 \dots Y_n}_{D_2} \quad Y_{n+1} \text{ unknown}$$

Goal: create $\hat{C}_n$ s.t.

$$\mathbb{P}(Y_{n+1} \in \hat{C}_n) \geq 1 - \alpha$$

Assume $Y_1, \dots, Y_n, Y_{n+1} \stackrel{\text{iid}}{\sim} P$

extend to exchangeability

the rank of Y_{n+1} is uniformly distributed over

the value of $Y_1, \dots, Y_n, Y_{n+1}$

↓
ranks

⇒

rank
→

$Y_1, \dots, Y_{n+1}$

$Y_{(1)}, \dots, Y_{(n)}, Y_{(n+1)}$

$$P(X_{n+1} \text{ is among } \lceil (1-\alpha)(n+1) \rceil \text{ smallest of } x_1, \dots, x_{n+1}) \geq 1-\alpha$$